WORKSHEET 4.2

Motion Along a Line

Unit 4: Derivatives in Context

Key ideas

- Velocity $v = x'$; acceleration $a = v' = x''$; speed $= |v|$.
- The particle is **at rest** when $v = 0$ and **changes direction** when v changes sign.
- Speed is increasing when v and a have the **same sign**, decreasing when they have opposite signs.
- For total distance, find the positions at the start, at each change of direction and at the end, and add the distances between them.

Worked examples

Example 1 · Velocity and rest

A particle has position $x(t) = t^3 - 12t^2 + 36t$ for $t \geq 0$. Find $v(t)$, $a(t)$ and the times at which it is at rest.

Solution. $v(t) = 3t^2 - 24t + 36 = 3(t-2)(t-6)$ and $a(t) = 6t - 24$.

At rest at $t = 2$ and $t = 6$.

Example 2 · Speeding up or slowing down

For the same particle, is the speed increasing or decreasing at $t = 3$?

Solution. $v(3) = -9$ and $a(3) = -6$.

Increasing: velocity and acceleration have the same sign.

Example 3 · Total distance

For the same particle, find the total distance travelled for $0 \leq t \leq 7$.

Solution. $x(0) = 0$, $x(2) = 32$, $x(6) = 0$, $x(7) = 7$. The distances are 32, 32 and 7.

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Practice

CORE

1. A particle has position $s(t) = t^2 - 8t + 5$. Find its velocity.

2. For the particle in Question 1, at what time is it at rest?

3. For the particle in Question 1, find its acceleration.

4. For the particle in Question 1, on what interval of $t \geq 0$ is it moving in the negative direction?

EXAM LEVEL

5. A particle has position $x(t) = t^3 - 3t^2$ for $t \geq 0$. At what time does it change direction? Justify your answer.

6. For the particle in Question 5, is the speed increasing or decreasing at $t = 1.5$? Give a reason.

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7. For the particle in Question 5, find the total distance travelled for $0 \leq t \leq 3$.

8. On a small moon, a ball thrown upward has height $h(t) = 24t - 4t^2$ metres after t seconds. Find its maximum height and its velocity when it returns to the ground.

Question 9 · Free-response practice

A particle moves along the x-axis with position $x(t) = 2t^3 - 9t^2 + 12t + 1$ for $t \geq 0$.

(a) Find $v(t)$ and $a(t)$.

(b) On what interval is the particle moving to the left? Justify your answer.

(c) Find the total distance travelled by the particle for $0 \leq t \leq 3$.

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Worked solutions

1. $v(t) = 2t - 8$.
2. $t = 4$.
3. $a(t) = 2$.
4. $0 \le t < 4$, where $v(t) < 0$.
5. $v(t) = 3t^2 - 6t = 3t(t-2)$ changes from negative to positive at $t = 2$.
6. $v(1.5) = -2.25$ and $a(1.5) = 6(1.5) - 6 = 3$: opposite signs, so the speed is decreasing.
7. $x(0) = 0$, $x(2) = -4$, $x(3) = 0$: $4 + 4 = 8$.
8. $h'(t) = 24 - 8t = 0$ at $t = 3$, so the maximum height is $h(3) = 36$ m. It lands at $t = 6$ with velocity -24 m/s.
9. (a) $v(t) = 6t^2 - 18t + 12$ and $a(t) = 12t - 18$.
   (b) $v(t) = 6(t-1)(t-2) < 0$ for $1 < t < 2$.
   (c) $x(0) = 1$, $x(1) = 6$, $x(2) = 5$, $x(3) = 10$: $5 + 1 + 5 = 11$.

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